SUMMER INTERNSHIP PRESENTATION

ON

VOCAL-LIBRA: A SMALL LANGUAGE MODEL FOR AUTOMATIC SPEECH RECOGNITION

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SUMMER INTERNSHIP CERTIFICATE

This is to certify that Mr. Raghavan Balanathan, student of National Institute of Technology, Trichy has under taken project Internship in the IIT Madras and worked under Dr. S. Umesh and successfully completed the project "Speech Technologies in Indian Languages" in the Department of Electrical Engineering, IIT Madras during the period 23rd May, 2024 to 26th July, 2024.

The performance of the student in Fellowship Programme was very good.

Head of the Department

ABSTRACT

The research aims to develop a novel multi-modal small language model (MSLM) that integrates text and speech modalities, enabling enhanced cross-modal conversational abilities and advancing the pursuit of Artificial General Intelligence (AGI).

By initializing the MLLM with pre-trained weights from a text-only language model and incorporating intermediate speech representations, the model utilizes extensive text data to improve speech processing capabilities.

A two-stage training strategy is proposed, involving modality-adaptation pretraining and cross-modal instruction fine-tuning, which strengthens the model's proficiency in both understanding and generating speech and text.

INTRODUCTION

Human communication heavily relies on speech, a dynamic and complex waveform that conveys critical information about the speaker's language, accent, gender, age, and emotional state.

Automatic Speech Recognition (ASR) is designed to translate spoken language into written text, which enables applications such as automated transcription, and real-time translation.

Traditional ASR systems consist of an acoustic model and a language model. The acoustic model maps audio features to speech units (like phonemes or words), while the language model captures linguistic patterns to improve the contextual accuracy of the transcription.

INTRODUCTION

Figure 1.1: Automatic Speech Recognition

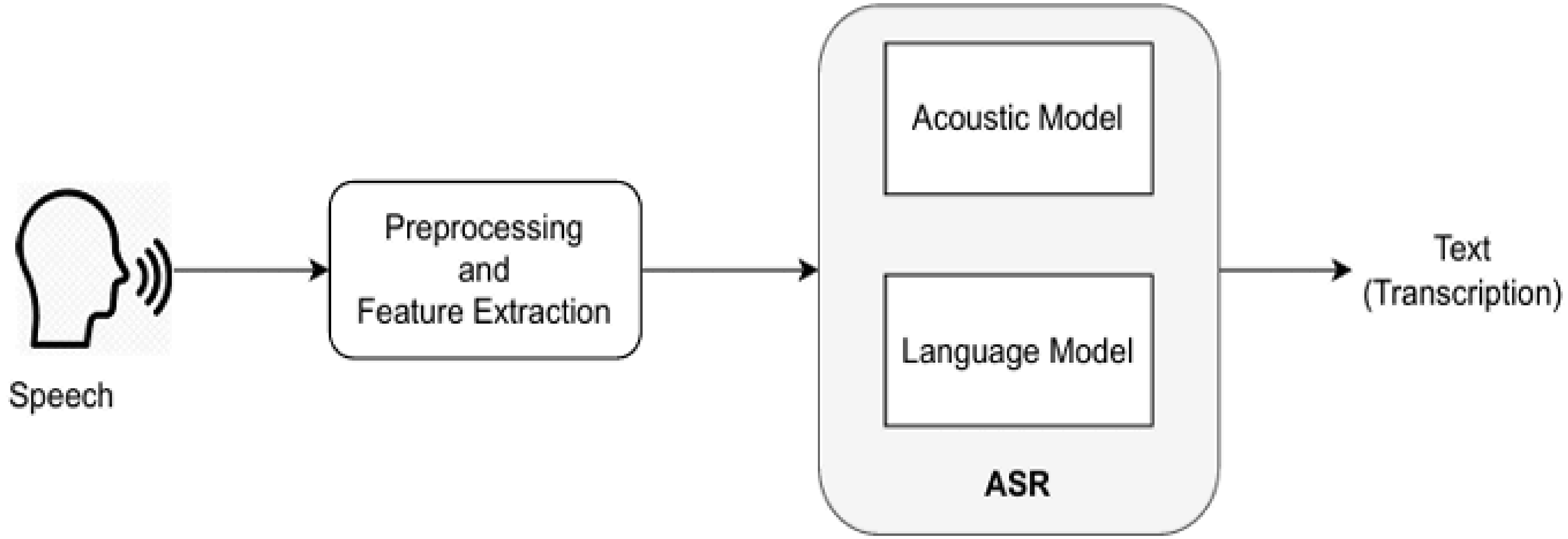


Figure 1.2: Traditional Automatic Speech Recognition

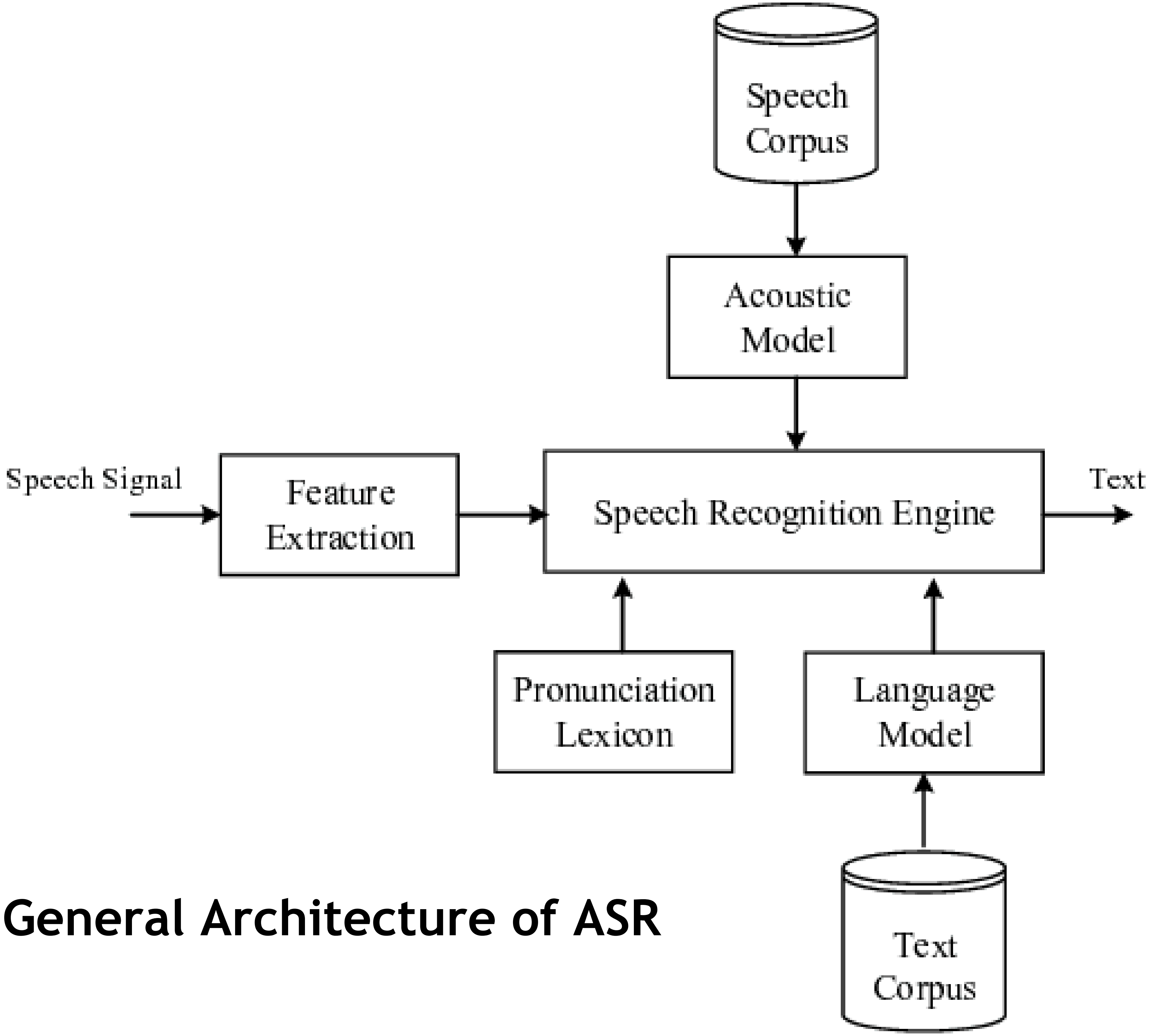
LITERATURE SURVEY

- *Evolution from Statistical to End-to-End Models:* Traditional ASR relied on HMMs for acoustics and n-gram models for language (Gales and Young, 2008; Katz, 1987). Deep learning-based end-to-end models now dominate, directly processing raw speech to text and improving robustness to variations in speech (Graves et al., 2006, 2013; Chan et al., 2016).

- Integrated Speech-Language Systems: The typical speech-language model connects LLMs with ASR or TTS models in a cascading setup. New research explores discrete token encoding of speech(Huang et al., 2023a; Baevski et al., 2020; Lakhotia et al., 2021).

- *Multimodal Language Models:* LLMs like GPT-4 and PALM-E are expanding into multimodal tasks, though they still struggle with handling continuous signals like audio (OpenAI, 2023; Driess et al., 2023; Liu et al., 2023).

LITERATURE SURVEY



General Architecture of ASR

IDENTIFIED GAPS

- *High Data Requirement*: ASR models need extensive labeled data, which is challenging to obtain for diverse languages and tasks.
- *High Computational Demand*: ASR models require substantial computational resources, limiting accessibility.
- *Limitations of LLM Cascade Models*: Cascaded ASR models using LLMs struggle with misaligned speech-text representations and fail to retain cues like emotion and prosody. [Shen et al., 2023; Huang et al., 2023a]

OBJECTIVE

1. Develop a pipeline to build a small language model for cross-modal conversational capabilities that can overcome the limitations of cross-modal knowledge transfer.
2. Extract acoustic features from HuBERT Base model to generate discrete and intermediate speech representation by training a K-Means model, suitable for language modelling.
3. Implement techniques to finetune pre-trained Phi-2 model for ASR effectively.
4. Evaluate the performance of the developed models on Libri Test sets.

DATASET

The LibriSpeech dataset (Panayotov et al., 2015) is a comprehensive, freely available corpus of English-language speech designed for training and testing automatic speech recognition (ASR) systems.

The dataset contains around 1,000 hours of 16 kHz spoken English text from audiobooks in the public domain sourced from the LibriVox project.

The dataset is organized as follows:

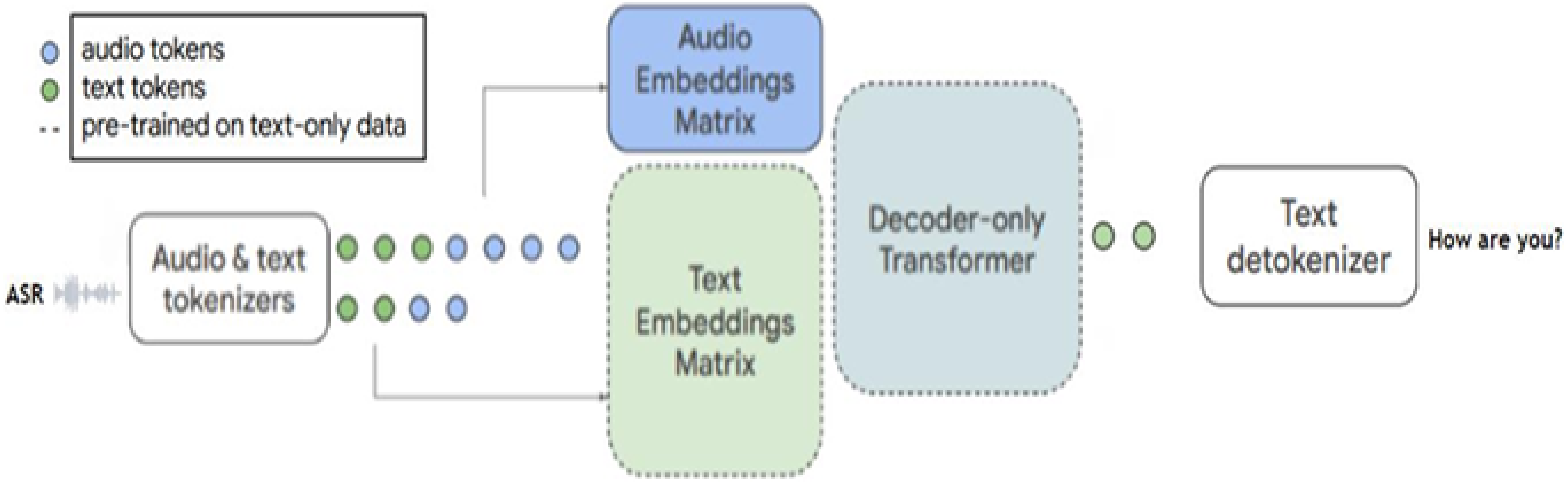
- Training sets: 460 hours ("clean") and 500 hours ("other")
- Validation sets: 5 hours each for "clean" and "other"
- Test sets: 5 hours each for "clean" and "other"

MODEL COMPONENTS

The discrete unit extractor utilizes the HuBERT Base (Hsu et al., 2021) to convert continuous speech signals into a series of discrete units. HuBERT is a self-supervised model that operates by predicting discrete labels for masked audio segments, utilizing k-means clustering on the model's intermediate representations. A k-means model subsequently transforms these representations into a sequence of cluster indices.

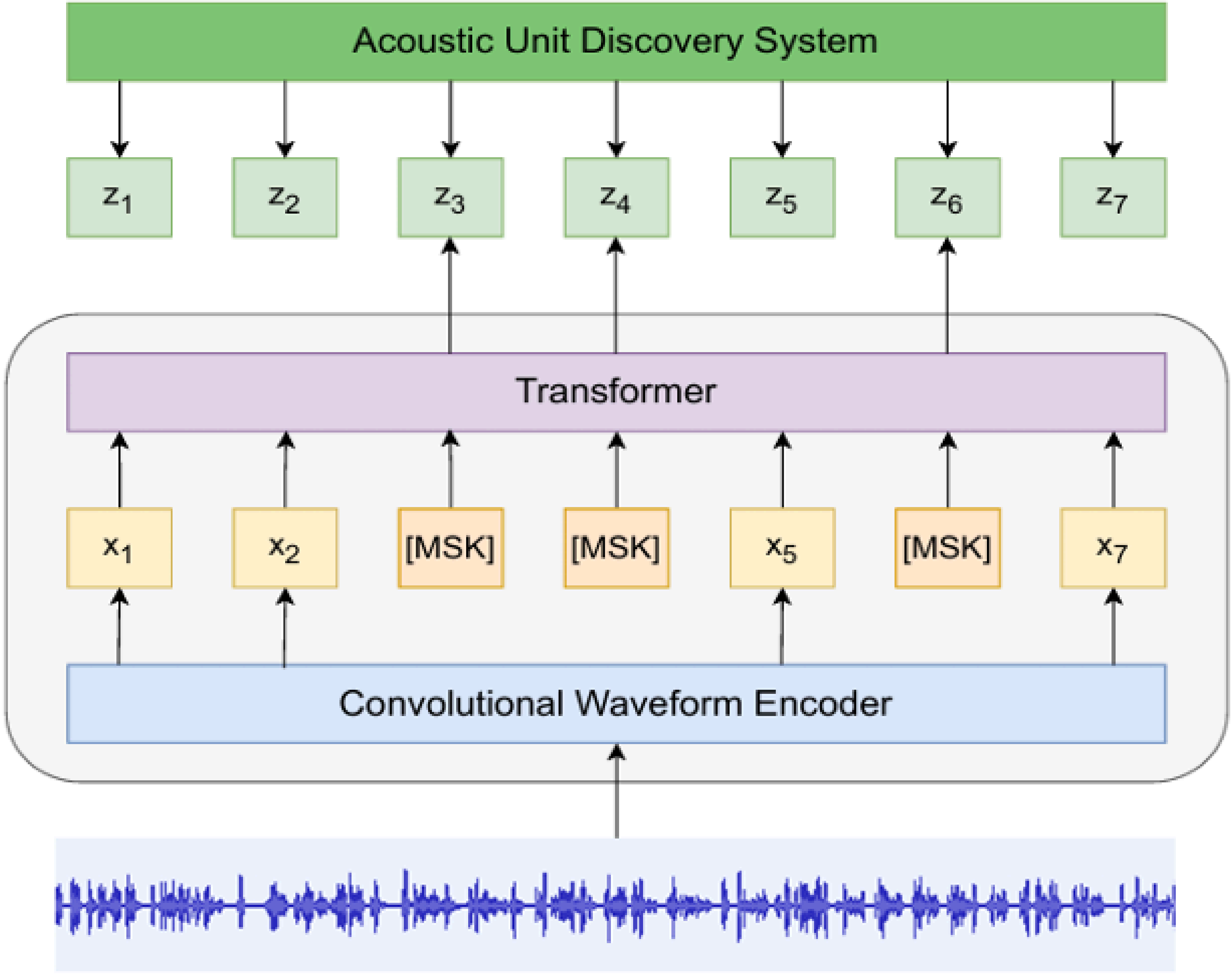
We use the Phi-2 model by Microsoft Research as our Small Language Model. Phi-2 which comprises an embedding layer, multiple transformer blocks, and an LM head layer, is a 2.7 billion-parameter language model that demonstrates outstanding reasoning and language understanding capabilities, showcasing state-of-the-art performance among base language models with less than 13 billion parameters.

BLOCK DIAGRAM



Proposed Model Structure

HuBERT Architecture



IMPLEMENTATION

To incorporate speech discrete representation into SLM, we expand the vocabulary and corresponding embedding matrix first.

We divide the training process into two stages.

1. The first stage is Modality Adaptation Pre-training on unpaired speech data.
2. The second stage is Cross-modal Instruction Fine-Tuning.

EXPANDING VOCABULARY

By sharing the same underlying architecture for both text and audio, the model benefits from cross-modal knowledge transfer. Additionally, the attention mechanisms in the decoder can seamlessly integrate information from both text and audio sequences. Thus, we can implement a simple adjustment to convert a text-only model into one that processes both text and audio by enlarging the embeddings matrix M to be of size $(t + a) \times e$, where a denotes the count of audio tokens

MODALITY ADAPTATION PRE-TRAINING

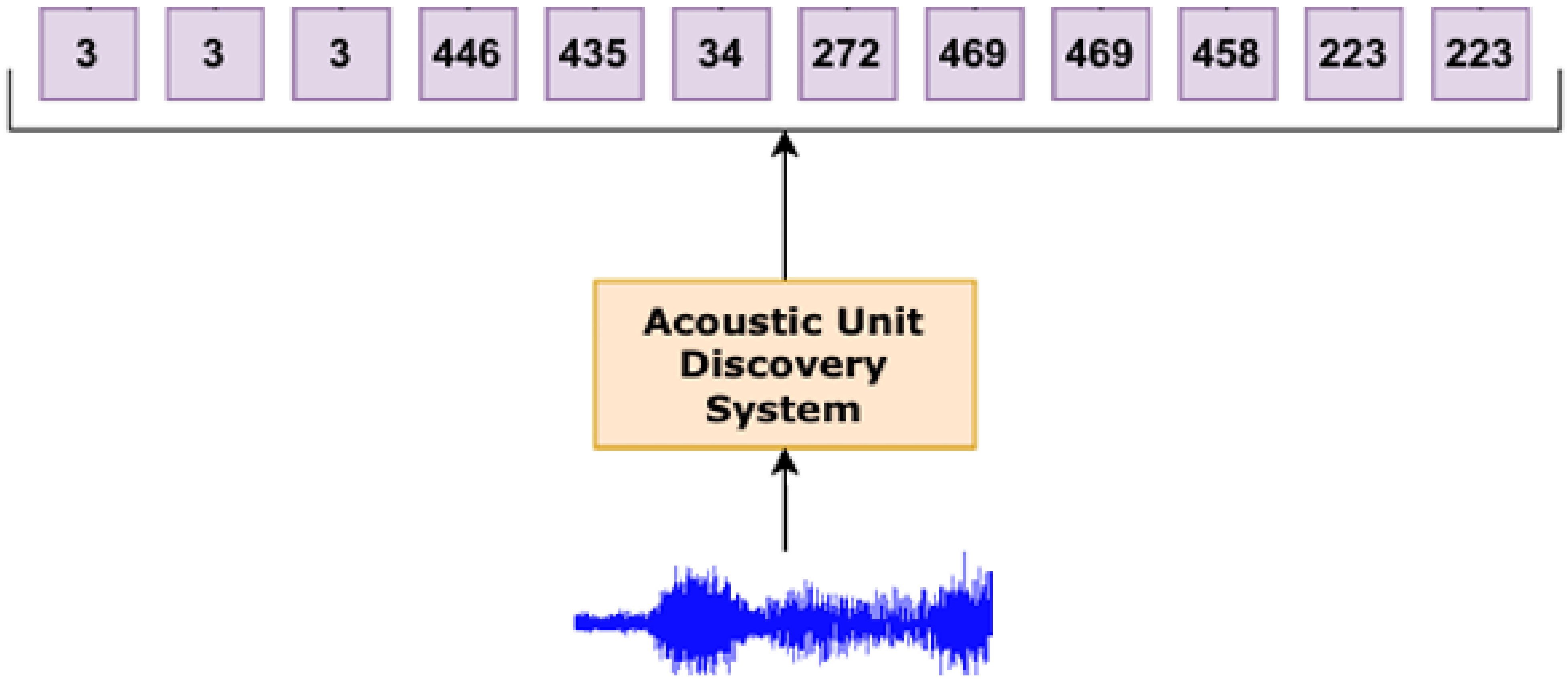
To enable LM to handle discrete speech units' modality, we utilize an unlabeled speech corpus to train LM in a next-token prediction task.

$$\mathcal{L}(L|C) = -\sum_{j=1}^{m}\sum_{i=1}^{n_j} \log P(u_{i,j}|u_{<i,j}; L)$$

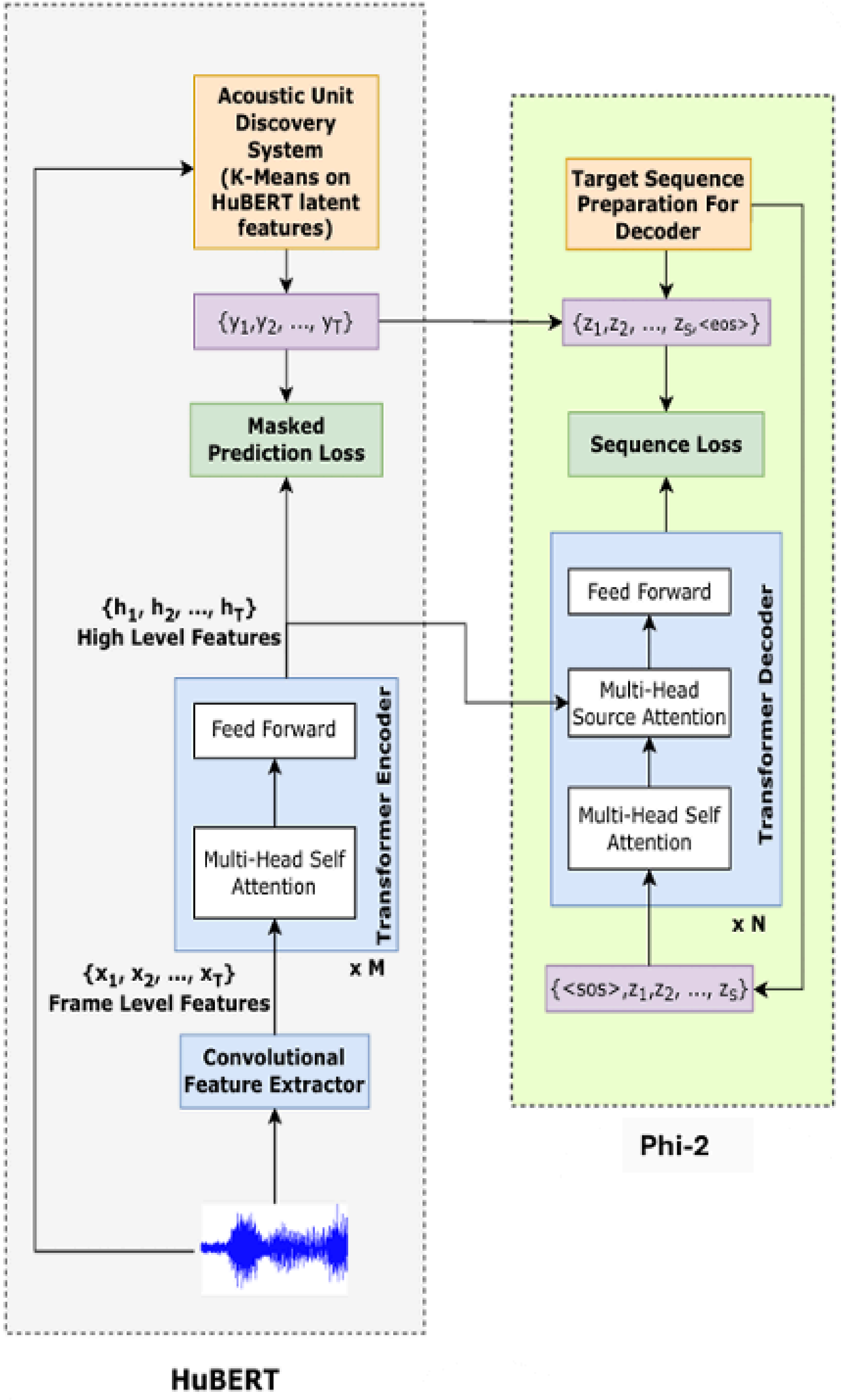
CROSS-MODAL INSTRUCTION FINE-TUNING

In this stage, we align speech and text modalities utilizing paired data. We use LibriSpeech dataset to map audio and its transcription for ASR to derive mix dataset I, which consists of samples T1, T2, . . . , Tx

BLOCK DIAGRAM



Discrete Speech Units Representation &
Model Architecture



EXPERIMENTAL SETUP

- The system configuration includes 8 NVIDIA DGX A100 40GB GPUs.
- For stage 1, we train for 1 epoch, 8000 steps with context size 2048. For stage 2, we train for 2 epochs, 4000 steps with context size 2048.
- We finetune with the Adafactor optimizer with a constant learning rate of 5e-5 and dropout rate of 0.1, and we use loss masking on the inputs.
- We used LibriSpeech 960 hours for training in both stages and performed inference and evaluation on test datasets. For decoding, we set the maximum sequence length to 2048 and set the temperature to 0.8. Training details are logged to Weights & Biases (Wandb).

EVALUATION METRIC

Word Error Rate (WER) is a widely used metric for evaluating the performance of Automatic Speech Recognition (ASR) systems. It measures the difference between the transcriptions generated by the ASR system and the ground truth transcriptions on the word level.

WER is calculated based on the number of substitutions (S), deletions (D), and insertions (I) that are required to transform the ASR system's output into the ground truth transcription. The formula for calculating WER is given by

$$\text{WER} = (S+D+I)/N$$

where N represents the total number of words in the ground truth transcription. The lower the WER, the better the performance of the ASR system.

RESULTS

Table 4.1: WER comparison with other ASR models on LibriSpeech benchmark

Model	Test-Clean WER	Test-Other WER	Extra Training Data
Local Prior Matching (Large Model) (Hsu et al., 2020)	7.19%	20.84%	✓
Deep Speech 2 (Amodei et al., 2015)	5.33%	13.25%	✗
CTC-CRF 4gram-LM (Xiang & Ou, 2019)	4.09%	10.65%	✗
Convolutional Speech Recognition (Zeghidour et al., 2018)	3.26%	10.47%	✓
MT4SSL (Ma et al., 2022)	3.4%	9.6%	✗
Proposed Model	**3.17%**	**7.82%**	✗

RESULTS

Table 4.2: WER comparison on training from scratch vs finetuning

Initial Checkpoint	Test-Clean WER	Test-Other WER
Phi-1.5 1.3B from scratch	10.14%	16.85%
Phi-2 2.7B from scratch	9.75%	15.22%
Phi-2 2.7B finetuned	3.17%	7.82%

TRAINING ANALYSIS

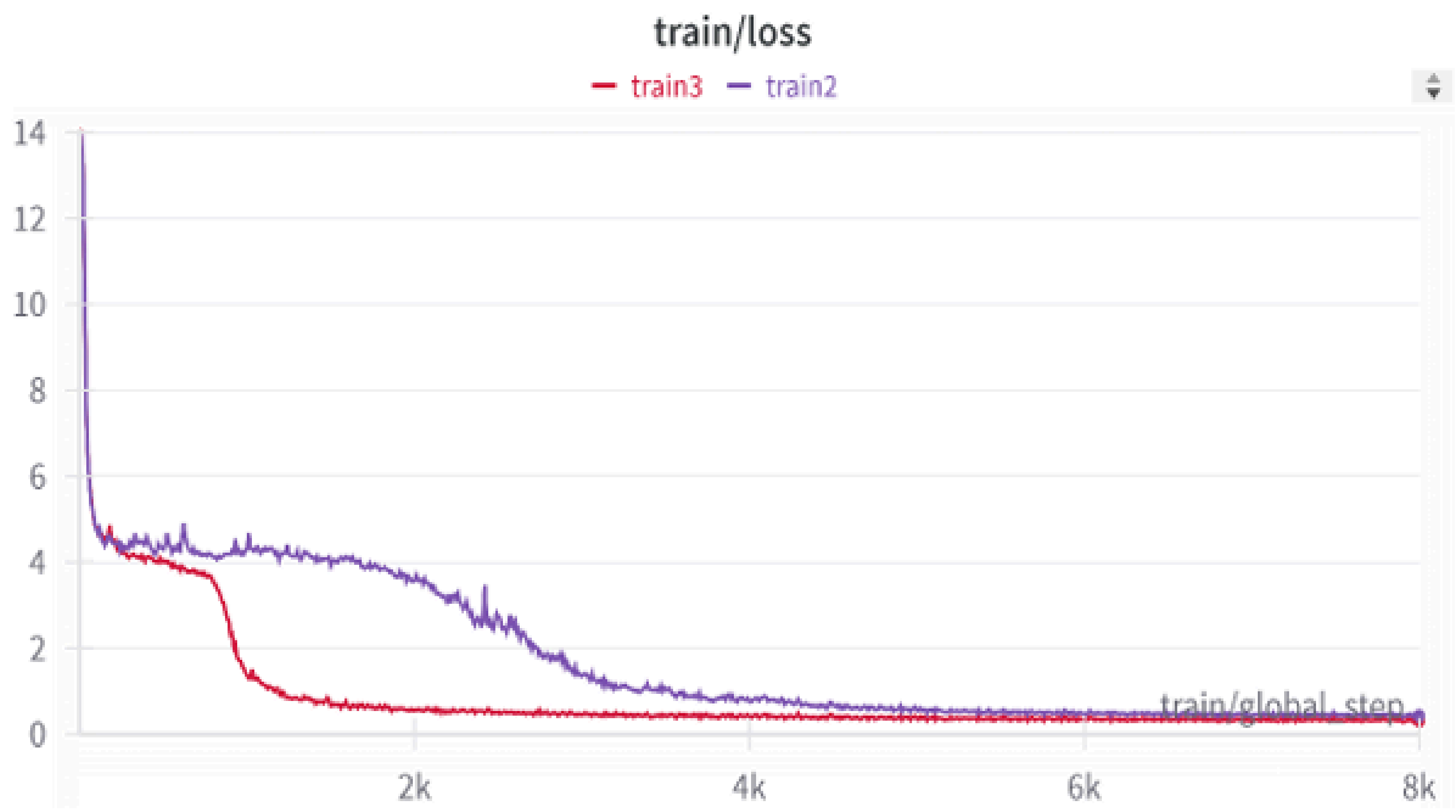


Figure 4.1: Training Loss curve for dual stage Modal Adaptation

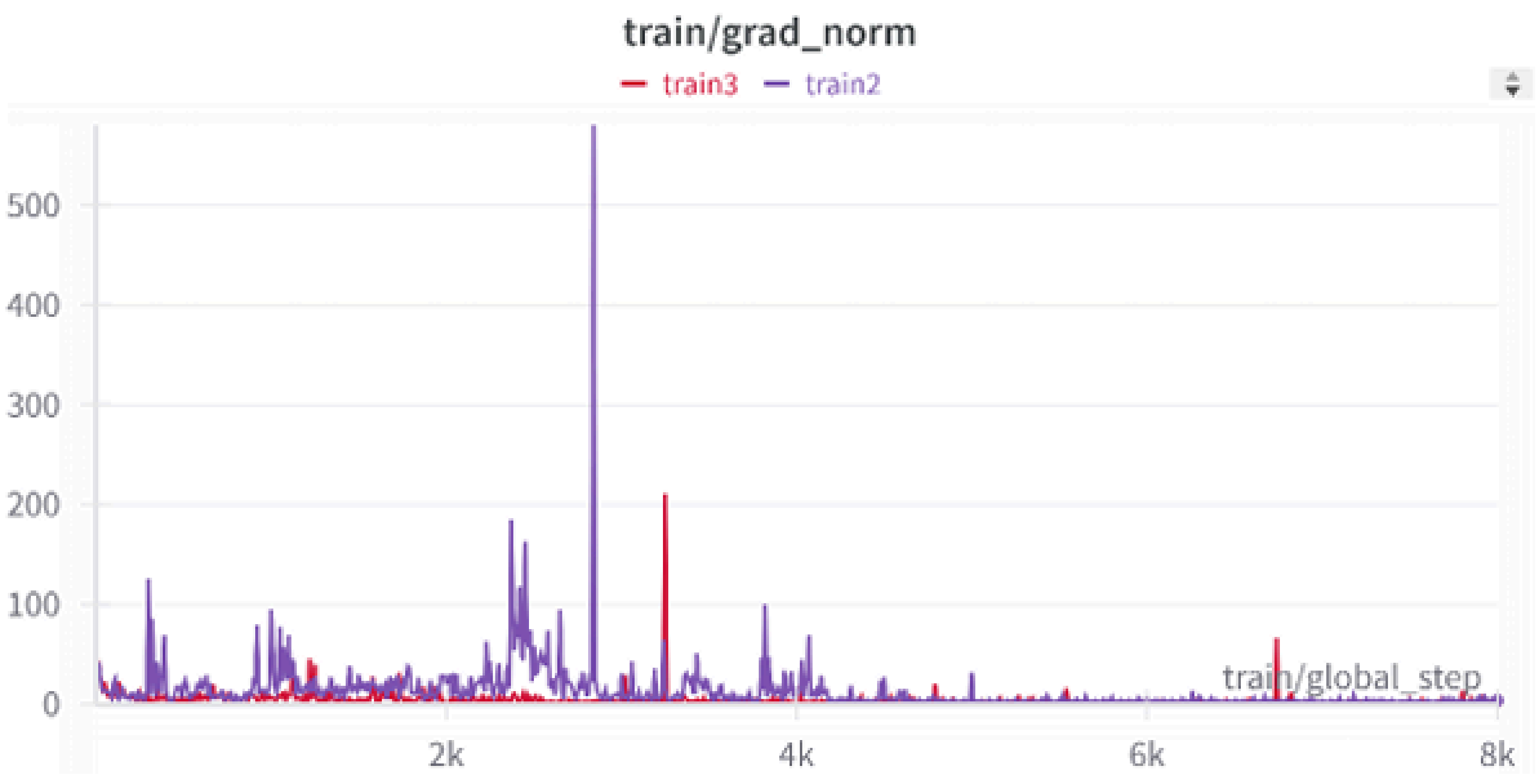


Figure 4.2: Gradient Norm Analysis during Training phases

TRANSCRIPTION ANALYSIS

Table 4.3: Analysis of Model Performance Across Various Transcription Challenges

Case Studies	Original Transcription	Model Transcription	WER (%)
Complex Sentence Transcription	If we're laying out a garden, planning one before the house, you know, and there you've a tree that's stood for centuries in the very spot. Old and gnarled it may be, and yet you don't cut down the old fellow to make room for the flowerbeds, but lay out your beds so as to take advantage of the tree.	If we are laying out to garden, planning one before the house, you know, and they you have a tree that stood a century in the very spot, old and gnarled it may be, and yet you don't cut down the old fellow to make room for the flower beds. But lay out your beds, so as to take advantage of the tree.	0.1094
	Oh, to shoot My soul's full meaning into future years, That they should lend it utterance, and salute Love that endures, from life that disappears.	Oh, to shoot my soul's full meaning into future years That they should lend it utterance, and salute Love that endures from life that disappears.	0.0
Named Entity Substitution Errors	Levin advanced, but utterly forgetting what he was to do, and much embarrassed, he turned to Sergey Ivanovitch with the question, Where am I to put it?	Levin advanced. But utterly forgetting what he was to do, and much embarrassed, he turned to Sir Jervis Evelyn with a question. Where am I to put it?	0.1481
	Father, thee's unjust to Philip. He's going into business.	Father, they's unjust to fill up. He's going into business.	0.3636
	Paul sticks to his theme.	Poultice to his theme.	0.4
Punctuation Integrity	There's one thing sure I'll not be suspected of complicity.	There's one thing sure I'll not be suspected of complicity.	0.0
	Why don't we cut down our parks for timber?	Why don't we cut down our box for tomorrow?	0.2
	There's a class instinct, too, of what one ought and oughtn't to do.	There's a class instinct, too, of what one ought and ought not to do.	0.0
Phonetic Confusion	To my thinking, I'd cut down that lime tree.	To mind thinking, I've got down that line, you see.	0.6
	Have I attempted to defraud you?	I've a tendency of the fraud you.	1.0
	Was Moses justified in resisting the Egyptian taskmaster?	Whose moose's its just findin' resistance in the Egyptian tat's master?	1.375

CONCLUSION

- Cross-Modal Integration: Vocal-Libra enhances speech-text interactions for advanced AI applications like virtual assistants.
- Hybrid Model Advancement: Combines speech and text, building on Phi-2's foundation to improve generative abilities.
- E2E Multimodal Understanding: Bridges speech and text by end-to-end design, leading the way toward more intuitive and context-aware AI systems.

FUTURE WORKS

- Expand Model Functionality: Integrate AST, S2ST, and TTS systems, especially for Indian languages.

- Enhance Multilingual Support: Include more languages, focusing on low-resource ones.

- Improve Audio Tokenization: Increase robustness to noise and better phonetic-semantic alignment.

- Incorporate Emotional Tone: Enable generation of responses with varied emotional tones and use unlabelled data effectively.

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THANK YOU!!!

ANY QUESTIONS?